AP CALCULUS AB

Unit 1 • Worksheet 1-6A

Worksheet 1-6A • Average Rate of Change

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the average rate of change of $f(x) = x^2 - 4x$ on $[1, 5]$?

(A) 2
(B) 8
(C) -2
(D) 4

Question 2

NO CALCULATOR

What is the average rate of change of $g(x) = x^3$ on $[-1, 2]$?

(A) 9
(B) 3
(C) $\frac{7}{3}$
(D) 1

Question 3

NO CALCULATOR

What is the average rate of change of $h(x) = \sqrt{x}$ on $[4, 9]$?

(A) 5
(B) 1
(C) $\frac{1}{5}$
(D) $\frac{3}{9}$

Question 4

NO CALCULATOR

t (h)	0	3	5	9
$P(t)$ (people)	120	150	210	300

What is the average rate of change of P on $3 \le t \le 9$?

(A) 150 people
(B) 30 people per hour
(C) 20 people per hour
(D) 25 people per hour

Question 5

NO CALCULATOR

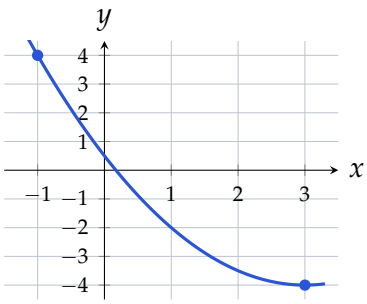
$C(x)$ is the cost, in FCFA, of x kilograms of rice. What are the units of $\dfrac{C(10)-C(4)}{10-4}$?

(A) FCFA
(B) FCFA per kilogram
(C) kilograms per FCFA
(D) kilograms

Question 6

NO CALCULATOR

The graph of f is shown. What is the average rate of change of f on $[-1,3]$?



(A) 2
(B) -8
(C) -2
(D) $-\frac{1}{2}$

Question 7

NO CALCULATOR

The average rate of change of f on $[0,4]$ is 3, and $f(0)=5$. What is $f(4)$?

(A) 17
(B) 12
(C) 8
(D) 7

Question 8

NO CALCULATOR

On a graph, the average rate of change on $[a,b]$ is the slope of

(A) the tangent line at a
(B) the x-axis
(C) the tangent line at b
(D) the secant line through $(a,f(a))$ and $(b,f(b))$

Question 9 (Free response) NO CALCULATOR

A tank is drained.

t (min)	0	10	20	30	40
$V(t)$ (L)	500	420	360	320	300

(a) Find the average rate of change of V on $[0, 40]$. Give units and interpret it.
(b) Find the average rate of change of V on $[20, 40]$.
(c) Find the average rates on $[0, 10]$, $[10, 20]$, $[20, 30]$, $[30, 40]$. Is the water leaving faster or slower over time?